

Fabric Management Dashboard

Developer Help & Technical Documentation

INR Global Sourcing, Tirupur — Garment Factory Sourcing

Version: 1.0.0

Built with: Laravel 11 + MySQL 8 + Blade + Tailwind CSS + Alpine.js + Chart.js

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Project Path: C:\Users\DELL\fabric-dashboard

1. Tech Stack

Component	Technology	Version / Notes
Backend Framework	Laravel	11.x (PHP 8.3+)
Language	PHP	8.3.33
Database	MySQL	8.4.9 (Server: 127.0.0.1:3306)
Database Name	fabric_db	DB user: <code>fabric</code> / pass: <code>fabric_pass_2026</code>
Frontend	Blade + Alpine.js	Alpine.js 3.x (via Vite)
CSS Framework	Tailwind CSS	3.x (via Vite + @tailwindcss/vite)
Charts	Chart.js	4.4.3 (CDN)
Auth	Laravel Breeze	2.4 (Blade stack)
Roles & Permissions	spatie/laravel-permission	8.3
Excel Import/Export	maatwebsite/excel	3.1 (PhpSpreadsheet wrapper)
PDF Generation	barryvdh/laravel-dompdf	3.1
Asset Bundler	Vite	8.x
Node.js	Node	v24.18.1 + npm 11.16
Session Driver	Database	Sessions stored in <code>sessions</code> table
Cache Store	Database	Cache stored in <code>cache</code> table
Queue Driver	Database	Jobs stored in <code>jobs</code> table

2. Default Login Credentials

IMPORTANT: These are seeded default accounts for local development. Change passwords immediately in production via User Management.

Role	Name	Email	Password	Permissions
ADMIN	System Admin	admin@fabricsourcing.in	Admin@12345	Full access: upload, edit, delete, manage users, resolve alerts, export
MANAGER	Factory Manager	manager@fabricsourcing.in	Manager@12345	Upload, edit, manage suppliers/buyers/styles, resolve alerts, export (no delete, no users)
VIEWER	Read Only Viewer	viewer@fabricsourcing.in	Viewer@12345	View dashboard, view records, export only (no edit, no upload, no resolve)

3. Role Permission Matrix

Action	Admin	Manager	Viewer
Upload data (Excel)	✓	✓	✗
Edit fabric/inspection records	✓	✓	✗
Delete records	✓	✗	✗
Manage users	✓	✗	✗
Manage suppliers/buyers/styles	✓	✓ (edit only)	✗
View dashboard	✓	✓	✓
Export reports	✓	✓	✓
Resolve alerts	✓	✓	✗

Implementation: Enforced via Spatie `role:admin|manager` middleware on routes + Laravel Policies (`FabricRecordPolicy` , `SupplierPolicy` , `UserPolicy` , `BuyerPolicy` , `StylePolicy` , `AlertPolicy`) + Blade `@can` / `@role` directives.

4. Project Folder Structure

```
fabric-dashboard/  
├─ app/  
│   ├─ Console/  
│   ├─ Http/  
│   │   ├─ Controllers/  
│   │   │   ├─ Auth/                (Breeze auth controllers)  
│   │   │   ├─ Admin/  
│   │   │   │   ├─ BuyerController.php  
│   │   │   │   ├─ FabricRecordController.php  
│   │   │   │   ├─ OverviewController.php  
│   │   │   │   ├─ StyleController.php  
│   │   │   │   ├─ SupplierController.php  
│   │   │   │   ├─ UploadController.php  
│   │   │   │   └─ UserController.php  
│   │   │   └─ AlertController.php  
│   │   │       ├─ Controller.php      (base, uses AuthorizesRequests)  
│   │   │       ├─ DashboardController.php  
│   │   │       └─ ProfileController.php  
│   │   └─ Requests/  
│   └─ Imports/  
│       └─ FabricImport.php          (Excel import with validation)  
├─ Exports/  
│   ├─ FabricRecordsExport.php      (filtered export)  
│   └─ FabricTemplateExport.php      (blank template download)  
├─ Models/  
│   ├─ Alert.php  
│   ├─ Buyer.php  
│   ├─ FabricRecord.php             (core record model)  
│   ├─ InspectionDetail.php  
│   ├─ KpiTarget.php  
│   ├─ QualityDefect.php  
│   ├─ Style.php  
│   ├─ Supplier.php  
│   ├─ UploadBatch.php  
│   └─ User.php                     (HasRoles trait)  
├─ Policies/  
│   ├─ AlertPolicy.php  
│   ├─ BuyerPolicy.php  
│   ├─ FabricRecordPolicy.php  
│   ├─ StylePolicy.php  
│   ├─ SupplierPolicy.php  
│   └─ UserPolicy.php  
├─ Services/  
│   ├─ AlertsEngineService.php       (auto-generates alerts)  
│   ├─ KpiService.php                (all 8 KPI formulas + statusColor)  
│   └─ SupplierRatingService.php      (nightly rating recalculation)  
├─ Providers/  
│   └─ AppServiceProvider.php         (registers policies)  
├─ bootstrap/  
│   └─ app.php                       (Spatie role/permission middleware)  
├─ config/  
│   └─ permission.php                (Spatie config)  
├─ database/
```

```
| └─ migrations/
|   | └─ 0001_01_01_000000_create_users_table.php
|   | └─ 0001_01_01_000001_create_cache_table.php
|   | └─ 0001_01_01_000002_create_jobs_table.php
|   | └─ 2026_08_10_075632_create_permission_tables.php
|   | └─ 2026_08_10_100001_create_buyers_table.php
|   | └─ 2026_08_10_100002_create_suppliers_table.php
|   | └─ 2026_08_10_100003_create_styles_table.php
|   | └─ 2026_08_10_100004_create_upload_batches_table.php
|   | └─ 2026_08_10_100005_create_fabric_records_table.php
|   | └─ 2026_08_10_100006_create_inspection_details_table.php
|   | └─ 2026_08_10_100007_create_quality_defects_table.php
|   | └─ 2026_08_10_100008_create_alerts_table.php
|   | └─ 2026_08_10_100009_create_kpi_targets_table.php
|   └─ seeders/
|       | └─ AdminUserSeeder.php          (3 default users)
|       | └─ DatabaseSeeder.php          (calls all seeders)
|       | └─ DemoDataSeeder.php          (sample buyers/styles/suppliers/records)
|       | └─ KpiTargetSeeder.php         (10 KPI targets)
|       | └─ RoleSeeder.php              (3 roles + 10 permissions)
└─ resources/
    | └─ css/app.css                    (Tailwind + print styles)
    | └─ js/app.js                      (Alpine.js bootstrap)
    └─ views/
        | └─ admin/
        |   | └─ buyers/index.blade.php
        |   | └─ fabric-records/{index,show,edit}.blade.php
        |   | └─ overview.blade.php
        |   | └─ styles/index.blade.php
        |   | └─ suppliers/{index,show}.blade.php
        |   | └─ upload/index.blade.php
        |   | └─ users/index.blade.php
        |   └─ auth/                    (Breeze login/register/reset)
        |   └─ components/
        |       | └─ confirm-modal.blade.php
        |       | └─ form-modal.blade.php
        |       | └─ kpi-card.blade.php    (colored KPI card)
        |       | └─ nav-item-link.blade.php (sidebar nav item)
        |       | └─ status-badge.blade.php (universal status badge)
        |       | └─ ... (Breeze components)
        |       └─ layouts/
        |           | └─ app.blade.php      (sidebar layout + Chart.js CDN)
        |           | └─ guest.blade.php
        |           | └─ navigation.blade.php
        |           └─ dashboard.blade.php  (5-row layout, 7 charts, alerts table)
        |           └─ profile/              (Breeze profile)
└─ routes/
    | └─ auth.php                      (Breeze auth routes)
    | └─ console.php                   (scheduled jobs: 06:00 + 06:05 daily)
    | └─ web.php                       (all application routes)
└─ .env                               (DB config: fabric_db / fabric / fabric_pass_2026)
└─ artisan
└─ composer.json
└─ package.json
└─ tailwind.config.js
└─ vite.config.js
```

5. Database Schema (10 Custom Tables)

5.1 users (extended from Breeze)

Column	Type	Notes
id	bigint PK	
name	varchar(100)	
email	varchar(150) unique	
password	varchar (hashed)	
role	enum(admin,manager,viewer)	default: viewer
is_active	boolean	default: true
last_login_at	timestamp nullable	updated on login
timestamps		

5.2 buyers

id, buyer_name (unique), contact_person, email, phone, is_active, timestamps

5.3 styles

id, style_number (unique), buyer_id (FK→buyers), order_quantity (decimal 12,2), target_date (date), status (enum: planning/in_progress/completed/on_hold), timestamps

5.4 suppliers

id, supplier_name (unique), mill_code, contact_person, phone, email, on_time_pct (decimal 5,2), quality_pct (decimal 5,2), rating (enum: excellent/good/average/poor), is_active, timestamps

5.5 fabric_records (core table)

id, record_date (date), buyer_id (FK), style_id (FK), supplier_id (FK), lot_no (varchar 50 unique), fabric_type, color, ordered_kg (decimal 12,2), received_kg (decimal 12,2), uploaded_by (FK→users), upload_batch_id (FK nullable), timestamps

5.6 inspection_details

id, fabric_record_id (FK cascade), inspected_kg, approved_kg, rejected_kg, gsm_actual, gsm_target, width_actual, width_target, pass_pct, bowing_pct, skewing_pct, shade_status (enum: approved/rejected/pending), inspected_by (FK→users), inspection_date, timestamps

5.7 quality_defects

id, fabric_record_id (FK cascade), defect_type, count (int), severity (enum: minor/major/critical), notes (text), timestamps

5.8 upload_batches (audit trail)

id, file_name, upload_type (enum: new_records/daily_update), uploaded_by (FK), total_rows, success_rows, error_rows, status (enum: validating/completed/failed), error_log (JSON), timestamps

5.9 alerts

id, fabric_record_id (FK nullable cascade), supplier_id (FK nullable cascade), alert_type (enum: delay/rejection/quality/shade), severity (enum: yellow/red), message (text), is_resolved (bool), resolved_by (FK), resolved_at, resolution_note (text), timestamps

5.10 kpi_targets (admin-editable targets)

kpi_key	target_value	comparison
inspection_completed	100	gte
pass_rate	98	gte
rejection_rate	2	lte
available_for_cutting	100	gte
shade_approval	100	gte
delayed_lots	0	lte
gsm_variation_pct	5	lte
width_variation_cm	1	lte
bowing_pct	3	lte
skewing_pct	3	lte

6. Routes Reference

6.1 Public / Auth Routes

Method	URI	Name	Description
GET	/login	login	Login page
POST	/login	login	Authenticate + set last_login_at
POST	/logout	logout	Logout
GET	/forgot-password	password.request	Forgot password form
GET	/reset-password/{token}	password.reset	Reset password form

6.2 Dashboard Routes (auth + verified)

Method	URI	Name	Description
GET	/dashboard	dashboard	Main dashboard (5 rows, 7 charts, alerts)
GET	/dashboard/data	dashboard.data	AJAX endpoint for filter updates (JSON)
PATCH	/alerts/{alert}/resolve	alerts.resolve	Resolve an alert (admin/manager)

6.3 Admin Routes (role: admin|manager, prefix /admin)

Method	URI	Name	Controller
GET	/admin	admin.overview	OverviewController@index
GET	/admin/upload	admin.upload.index	UploadController@index
POST	/admin/upload/validate	admin.upload.validate	UploadController@validateFile
POST	/admin/upload/import	admin.upload.import	UploadController@import
GET	/admin/upload/template	admin.upload.template	Download .xlsx template
GET	/admin/fabric-records	admin.fabric-records.index	FabricRecordController@index
GET	/admin/fabric-records-export	admin.fabric-records.export	Export filtered records to .xlsx
GET	/admin/fabric-records/{id}	admin.fabric-records.show	Detail view + quality metrics
GET	/admin/fabric-records/{id}/edit	admin.fabric-records.edit	Edit form (admin/manager)
PUT	/admin/fabric-records/{id}	admin.fabric-records.update	Update + recalc KPIs + alert scan
DELETE	/admin/fabric-records/{id}	admin.fabric-records.destroy	Delete (admin only, cascade)
GET/POST/PUT/DELETE	/admin/suppliers/*	admin.suppliers.*	SupplierController (resource)
PATCH	/admin/suppliers/{id}/toggle-active	admin.suppliers.toggle	AJAX toggle active
GET/POST/PUT/DELETE	/admin/buyers/*	admin.buyers.*	BuyerController (resource)

PATCH	/admin/buyers/{id}/toggle-active	admin.buyers.toggle	AJAX toggle active
GET/POST/PUT/DELETE	/admin/styles/*	admin.styles.*	StyleController (resource)

6.4 User Management Routes (role: admin only)

Method	URI	Name
GET	/admin/users	admin.users.index
POST	/admin/users	admin.users.store
PUT	/admin/users/{user}	admin.users.update
DELETE	/admin/users/{user}	admin.users.destroy
POST	/admin/users/{user}/reset-password	admin.users.reset-password
PATCH	/admin/users/{user}/deactivate	admin.users.deactivate

7. KPI Calculation Formulas

All implemented in `app/Services/KpiService.php` method `calculate(array $filters)`.

#	KPI	Formula	Target
1	Total Fabric Required	<code>SUM(fabric_records.ordered_kg)</code>	—
2	Total Fabric Received	<code>SUM(fabric_records.received_kg)</code>	—
3	Total Approved	<code>SUM(inspection_details.approved_kg)</code>	—
4	Inspection Completed %	<code>(SUM(inspected_kg) / SUM(received_kg)) × 100</code>	≥100%
5	Fabric Pass Rate %	<code>(SUM(approved_kg) / SUM(inspected_kg)) × 100</code>	≥98%
6	Rejection Rate %	<code>(SUM(rejected_kg) / SUM(inspected_kg)) × 100</code>	≤2%
7	Available for Cutting %	<code>(SUM(approved_kg) / SUM(ordered_kg)) × 100</code>	≥100%
8	Shade Approval %	<code>(COUNT(shade=approved) / COUNT(all)) × 100</code>	≥100%
9	Delayed Lots	<code>COUNT(received<ordered AND target_date<today AND status!=completed)</code>	=0

Quality Metrics (per lot)

- **GSM Variation %** = `ABS(gsm_actual - gsm_target) / gsm_target × 100` → flag if >5%
- **Width Variation (cm)** = `ABS(width_actual - width_target)` → flag if >1cm
- **Bowing % / Skewing %** → flag if >3%
- **Hole Defects** = `SUM(quality_defects.count WHERE defect_type='Hole')` → target 0

Status Color Coding

Color	Condition
GREEN	KPI meets or exceeds target
YELLOW	Within 10% of target but not met (e.g. pass rate 96–97.9% when target is 98%)
RED	More than 10% off target, OR "lower is better" KPI exceeding target

Helper: `KpiService::statusColor($value, $target, $comparison)` — single reusable method, consumed by `<x-kpi-card>` and `<x-status-badge>` Blade components.

8. Alerts Engine Rules

Implemented in `app/Services/AlertsEngineService.php`. Runs on every import AND nightly via scheduler at 06:05.

Condition	Type	Severity
-----------	------	----------

received_kg < ordered_kg AND target_date < today	delay	RED
target_date - today ≤ 3 days AND received_kg < ordered_kg	delay	YELLOW
lot rejected_kg / inspected_kg > 2%	rejection	RED
lot pass_pct < 96%	quality	YELLOW
shade_status = 'rejected'	shade	RED
supplier quality_pct < 90 after recalculation	quality	RED

9. Supplier Rating Formula

Implemented in `app/Services/SupplierRatingService.php` . Runs nightly at 06:00 + after each import.

- **On-Time** % = (lots delivered on/before target_date / total lots) × 100
- **Quality** % = (SUM approved_kg / SUM inspected_kg) × 100 (scoped to supplier)
- **Rating:**
 - quality ≥ 98 AND on_time ≥ 95 → excellent
 - quality ≥ 95 AND on_time ≥ 90 → good
 - quality ≥ 90 AND on_time ≥ 80 → average
 - else → poor

10. Dashboard Layout (5 Rows)

Row	Content	Charts
Filter Bar	Sticky top: Buyer, Style, Supplier, Fabric Type, Color, Date range + Apply / Clear / Export PDF buttons	—
Row 1	6 Executive Summary Cards (colored left border): Required, Received, Approved, Pass%, Reject%, Delayed Lots	—
Row 2	Trends & Overview	Line: Daily Receipt Trend (30d) + Donut: Fabric Status Breakdown
Row 3	Performance Analysis	Horizontal Bar: Supplier Performance (top 10) + Doughnut Gauge: Inspection Completion %
Row 4	Quality & Inventory	Pareto: Top Defects (bar + cumulative line) + Stacked Bar: Stock by Fabric Type
Row 5	Action Items	Bar+Line: Consumption vs Plan (by style) + Critical Alerts Table (with Resolve buttons)

Live filtering: All charts/cards update via AJAX fetch to `/dashboard/data` — no full page reload. Powered by Alpine.js + Chart.js.

11. Excel Upload / Import

Template Columns (in exact order)

#	Column	Required	Validation
1	Date	Yes	Must be YYYY-MM-DD format
2	Buyer	Yes	Auto-creates if new
3	Style	Yes	Auto-creates if new
4	Supplier	Yes	Auto-creates if new
5	Lot No	Yes	Unique (New) or must exist (Update)
6	Fabric Type	Yes	
7	Color	Yes	
8	Ordered Kg	Yes	Numeric only (no commas/currency)
9	Received Kg	Yes	Numeric only
10	Inspected Kg	No	Numeric
11	Approved Kg	No	Numeric
12	Rejected Kg	No	Numeric

13	GSM	No	Numeric
14	Width	No	Numeric
15	Pass %	No	Numeric
16	Defect Type	No	Text (e.g. Hole, Stain, Slub)
17	Defect Count	No	Integer
18	Severity	No	minor / major / critical

Flow: Download Template → Fill data → Choose Upload Type (New/Daily Update) → Choose File → **Validate** (shows green/red preview) → **Import** (creates upload_batch, upserts records in transaction, triggers SupplierRating recalc + AlertsEngine scan).

12. Scheduled Jobs

Defined in `routes/console.php` . Run `php artisan schedule:run` or set up a cron entry: `* * * * * cd /path-to-project && php artisan schedule:run >> /dev/null 2>&1`

Time	Job	Description
06:00 daily	recalculate-supplier-ratings	Recalculates on_time_pct, quality_pct, rating for all suppliers
06:05 daily	alerts-engine-scan	Scans all fabric records and generates delay/rejection/quality/shade alerts

13. How to Run the Application

13.1 Start the Development Server

```
cd C:\Users\DELL\fabric-dashboard
php artisan serve
<!-- Server runs at http://127.0.0.1:8000 -->
```

13.2 Build Frontend Assets (for production)

```
npm run build
```

13.3 Run in Dev Mode (hot reload)

```
npm run dev    <!-- in one terminal -->
php artisan serve  <!-- in another terminal -->
```

13.4 Reset Database (fresh migrate + seed)

```
php artisan migrate:fresh --seed --force
```

13.5 Run Scheduled Jobs Manually

```
php artisan schedule:run
```

13.6 Clear All Caches

```
php artisan config:clear
php artisan cache:clear
php artisan route:clear
php artisan view:clear
php artisan optimize:clear
```

13.7 Generate this PDF

```
php artisan docs:pdf
```

14. Environment Configuration (.env)

Key	Value
APP_NAME	Fabric Management Dashboard

APP_ENV	local
APP_DEBUG	true
APP_URL	http://localhost
DB_CONNECTION	mysql
DB_HOST	127.0.0.1
DB_PORT	3306
DB_DATABASE	fabric_db
DB_USERNAME	fabric
DB_PASSWORD	fabric_pass_2026
SESSION_DRIVER	database
QUEUE_CONNECTION	database
CACHE_STORE	database
MAIL_MAILER	log (password reset links stored in log file)

15. Key Files Quick Reference

File	Purpose
app/Services/KpiService.php	All 8 KPI formulas + statusColor() helper + chart data methods
app/Services/AlertsEngineService.php	6 alert rules, auto-generates alerts, skip duplicates
app/Services/SupplierRatingService.php	Recalculates supplier on_time/quality/rating
app/Imports/FabricImport.php	Excel import with row validation, auto-creates buyers/styles/suppliers
app/Exports/FabricRecordsExport.php	Filtered Excel export of fabric records
app/Exports/FabricTemplateExport.php	Blank template download with headers + sample row
routes/web.php	All application routes with role middleware
routes/console.php	Scheduled jobs (06:00 + 06:05 daily)
bootstrap/app.php	Spatie role/permission middleware aliases
app/Providers/AppServiceProvider.php	Policy registrations
resources/views/dashboard.blade.php	Main dashboard with 5 rows, 7 charts, filter bar, alerts
resources/views/layouts/app.blade.php	Sidebar layout with role-based nav items + Chart.js CDN
resources/views/components/kpi-card.blade.php	Reusable colored KPI card
resources/views/components/status-badge.blade.php	Universal status badge (green/yellow/red + all enum values)
resources/views/components/confirm-modal.blade.php	Reusable confirmation modal with form
resources/views/components/form-modal.blade.php	Reusable form modal (for add/edit)

16. Demo Data Seeded

Entity	Count	Details
Users	3	1 Admin + 1 Manager + 1 Viewer
Buyers	3	INR Global Sourcing, Prime Apparel UK, Nordic Textiles AB
Styles	4	STY-1001 to STY-1004 (various statuses)
Suppliers	4	Tirupur Mills, Coimbatore Fabrics, Karur Textiles, Erode Knit Fab
Fabric Records	8	Lots LOT-2401-001 to LOT-2403-008 across all styles
Inspection Details	8	1 per fabric record (some with rejections/shade issues)
Quality Defects	~10	Random defects (Hole, Stain, Slub, Misprint, etc.)
Alerts	~11	Auto-generated by AlertsEngine (delay, rejection, quality, shade)

KPI Targets	10	All targets from spec section 1.10
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17. Acceptance Checklist

- ✓ Every button in spec is wired to a real route (61 routes total)
- ✓ Every KPI formula matches KpiService implementation exactly
- ✓ Role restrictions enforced in routes (middleware) AND Blade (@can/@role)
- ✓ Excel upload rejects bad dates, currency symbols, commas, duplicate lots
- ✓ Dashboard filters (Buyer/Style/Supplier/Type/Color/Date) affect all 5 rows live via AJAX
- ✓ Color coding (green/yellow/red) consistent via shared statusColor() helper
- ✓ Alerts generated automatically on import AND nightly via scheduler
- ✓ `php artisan migrate:fresh --seed` runs with zero errors
- ✓ All 12 main endpoints return HTTP 200 when authenticated as admin
- ✓ Manager gets 403 on user management; Viewer gets 403 on upload + admin panel
- ✓ Excel template download + filtered export both produce valid .xlsx files
- ✓ Alert resolution saves resolution_note + resolved_at + resolved_by
- ✓ Supplier ratings auto-calculated nightly + on import
- ✓ Style status auto-flips to 'completed' when approved_kg ≥ order_quantity